

MitoChime

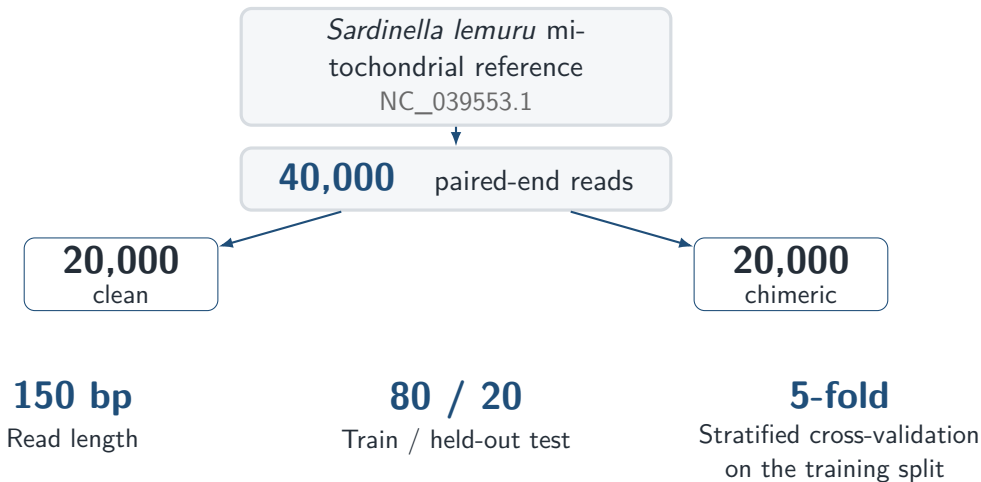
A Machine Learning Pipeline for Detecting PCR-Induced Chimeras in Mitochondrial Illumina Reads

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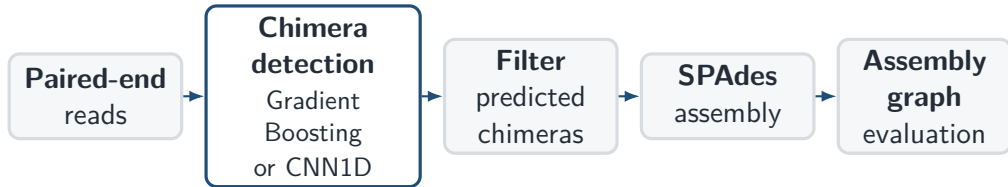
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NiDS2026

Experimental Setup



MitoChime Workflow



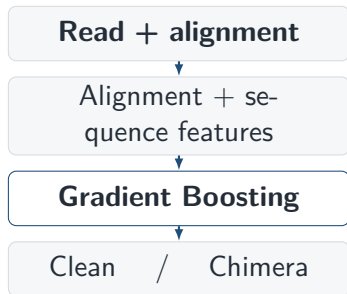
Predicted chimeric reads are removed **before assembly**.

Downstream impact is evaluated through the resulting assembly graph structure.

Two Approaches to Chimera Detection

Gradient Boosting

Feature-based



CNN1D

Sequence-based

